ANDRÉS GÓMEZ RAMÍREZ

Senior DevOps Engineer | Cloud & Infrastructure | AWS · Kubernetes · Terraform · IaC

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PROFILE

Senior DevOps and infrastructure engineer with 15+ years of experience building, automating, and operating large-scale Linux environments in both cloud and HPC contexts. Five years at Edenia delivering AWS infrastructure, Kubernetes clusters, Terraform IaC, GitHub Actions CI/CD, and Grafana/Prometheus observability for blockchain-based products. Seven years embedded in the ALICE experiment at CERN, contracted through Goethe University Frankfurt, managing critical cluster infrastructure using Ansible, Puppet, Zabbix, Vagrant, and OpenStack at scale. PhD in Cybersecurity with a research background in AI-driven monitoring and anomaly detection.

CORE SKILLS

Cloud & Infra	OpenStack, AWS (EC2, EKS, S3, CloudWatch, IAM, VPC), multi-account environments
Containers & Orchestration	Docker, Kubernetes (EKS), Helm, containerized microservices deployment
CI/CD	GitLab CI (pipelines, downstream, multi-project), GitHub Actions, Jenkins
IaC & Config Mgmt	Terraform, Puppet, Ansible, Vagrant — reproducible, version-controlled infrastructure
Monitoring & Observability	Grafana, Prometheus, Zabbix, AWS CloudWatch, ELK Stack — dashboards, alerting, anomaly detection
Scripting & Languages	Python, Bash/Shell, JavaScript/Node.js, Perl, C/C++
Security	CIS hardening, IAM least privilege, SSH key management, MFA, vulnerability scanning, CVE research
Spoken Languages	Spanish (Native), English (Fluent), German (Basic)

PROFESSIONAL EXPERIENCE

Blockchain DevOps Engineer — *Edenia* **2020 – 2025**

- Designed, deployed, and maintained Linux-based on-premise and cloud infrastructure on AWS (EC2, EKS, S3, IAM, VPC) for blockchain-focused products, managing environments across development, staging, and production.
- Provisioned and managed Kubernetes clusters, handling deployments, scaling, service mesh configuration, and rollout strategies for containerized microservices.
- Authored and maintained Terraform modules for infrastructure-as-code, enabling reproducible, version-controlled provisioning of all cloud resources.
- Built and maintained CI/CD pipelines using GitHub Actions, automating build, test, and deployment workflows across multiple projects.
- Implemented observability solutions using Grafana and Prometheus — designed dashboards, defined alerting rules, and tuned thresholds for reliable operational visibility.
- Worked within a distributed, remote-first team delivering infrastructure for blockchain ecosystems including EOSIO/Antelope, Ethereum, and Bitcoin networks.

DevOps Engineer — *CERN / ALICE Experiment (contracted via Goethe University Frankfurt)* **2013 – 2020**

- Built and maintained the security monitoring and infrastructure management system for the ALICE High-Level Trigger (HLT) computing cluster — a large-scale, heterogeneous Linux environment processing live detector data at the Large Hadron Collider.

- Managed infrastructure-as-code and configuration management for the ALICE HLT cluster using Puppet and Ansible, enforcing consistent hardened baselines across hundreds of compute nodes.
- Operated and monitored cluster health and performance using Zabbix, building high-signal dashboards and alert rules; contributed to the HLT read-out upgrade for LHC Run 2 (published in JINST).
- Worked with Vagrant and OpenStack for virtual environment provisioning, automating the lifecycle of development and test environments.
- Collaborated with physicists, engineers, and CERN security teams across a globally distributed organisation, delivering infrastructure that supported live scientific data-taking runs.

Co-Founder & CEO — *Sakundi ÖU* **2022 – 2026**

- Lead technical strategy and infrastructure for a security and digital-identity company, delivering DevOps and security services to enterprise clients.
- Design and maintain reproducible, auditable deployment pipelines and infrastructure-as-code for client environments.

DevOps & Security Developer — *Modulo Labs* **2025 – 2026**

- Develop and review cryptographic and smart contract code; manage deployment pipelines and infrastructure for a privacy-enhancing protocol on the Polygon ecosystem.

Part-Time Professor, Cybersecurity — *Costa Rica Institute of Technology (TEC)* **2026 – Present**

- Teach infrastructure security and DevOps practices within the Cybersecurity Master's programme.

Security Researcher — *Fluid Attacks* **2011 – 2013**

- Performed infrastructure assessments, source code audits, and penetration tests for banks, software companies, and ISPs. Identified and disclosed 6 CVEs.

SELECTED PROJECTS

Tikuna: AI-based monitoring system for the Ethereum P2P network — anomaly detection pipeline over graph-structured network data, funded by Ethereum Foundation Academic Research grant. github.com/sakundi/tikuna

Arhuaco: PhD research at CERN / Goethe University Frankfurt: Python/Perl framework implementing deep learning and isolation-based intrusion detection for the WLCG; published in Journal of Grid Computing. github.com/kuronosec/arhuaco

Zikuani: Privacy-preserving identity verification using zero-knowledge proofs. Cloud deployment infrastructure and CI/CD pipeline. DIF 2025 hackathon winner. github.com/kuronosec/zikuani

EDUCATION

PhD, Computer Science (Cybersecurity) — *Goethe University Frankfurt, Germany* **2013 – 2019**

Thesis: "Deep Learning and Isolation Based Security for Intrusion Detection and Prevention in Grid Computing"

Research conducted in collaboration with CERN, embedded in the ALICE experiment.

BSc, Computer Science — *Universidad de Antioquia, Medellín, Colombia* **2003 – 2011**

SELECTED PUBLICATIONS

- Gómez Ramírez, A., Al Sardy, L., Gómez Ramírez, F. — "Tikuna: An Ethereum Blockchain Network Security Monitoring System." ISPEC 2023, Springer LNCS vol. 14341.
- Gómez Ramírez, A., Lara, C., Betev, L., Bilanovic, D., Kebschull, U. (ALICE Collaboration) — "Arhuaco: Deep Learning and Isolation Based Security for Distributed High-Throughput Computing." Journal of Grid Computing.
- Gómez Ramírez, A., Martínez Pedreira, M., Grigoras, C., Betev, L., Lara, C., Kebschull, U. (ALICE Collaboration) — "A Security Monitoring Framework for Virtualization Based HEP Infrastructures." Journal of Physics: Conference Series, 898(10), 2017.

- Gómez Ramírez, A., Lara, C., Kebschull, U. (ALICE Collaboration) — "Intrusion Prevention and Detection in Grid Computing — The ALICE Case." Journal of Physics: Conference Series, 664(6), 2015.
- Engel, H., Alt, T., Breitner, T., Gómez Ramírez, A. et al. (ALICE Collaboration) — "The ALICE High-Level Trigger Read-Out Upgrade for LHC Run 2." JINST, IOP Publishing, 2015.
- The ALICE Collaboration (incl. Gómez Ramírez, A.) — "O2: A Novel Combined Online and Offline Computing System for the ALICE Experiment." Journal of Physics: Conference Series, 513(1), 2014.
- *6 CVEs assigned including MS13-056 (Microsoft DirectShow).*
- *Full publication list at https://orcid.org/0000-0003-1529-4614.*